

Parity-Check Polynomial

- An (n, k) cyclic code is uniquely specified by another polynomial of degree k , which is called *parity-check polynomial* defined by

$$h(X) = 1 + \sum_{i=1}^{k-1} h_i X^i + X^k, \text{ where } h_i \text{ is 0 or 1.}$$

- Here $h_0 = h_k = 1$.
- The generator and parity-check polynomials are equivalent to the generator ($\mathbf{G}$) and the parity-check ($\mathbf{H}$) matrices.
- Equivalent to $\mathbf{H}\mathbf{G}^T = 0$, it has: $g(X)h(X) \bmod (X^n + 1) = 0$.
- Accordingly, we may state that the generator and parity-check polynomials are factors of $X^n + 1$ as shown below.

$$g(X)h(X) = X^n + 1$$

A (7, 4) CYCLIC CODE GENERATED BY $g(X) = 1 + X + X^3$

Messages	Code Vectors	Code polynomials
(0 0 0 0)	0 0 0 0 0 0 0	$0 = 0 \cdot g(X)$
(1 0 0 0)	1 1 0 1 0 0 0	$1 + X + X^3 = 1 \cdot g(X)$
(0 1 0 0)	0 1 1 0 1 0 0	$X + X^2 + X^4 = X \cdot g(X)$
(1 1 0 0)	1 0 1 1 1 0 0	$1 + X^2 + X^3 + X^4 = (1 + X) \cdot g(X)$
(0 0 1 0)	0 0 1 1 0 1 0	$X^2 + X^3 + X^5 = X^2 \cdot g(X)$
(1 0 1 0)	1 1 1 0 0 1 0	$1 + X + X^2 + X^5 = (1 + X^2) \cdot g(X)$
(0 1 1 0)	0 1 0 1 1 1 0	$X + X^3 + X^4 + X^5 = (X + X^2) \cdot g(X)$
(1 1 1 0)	1 0 0 0 1 1 0	$1 + X^4 + X^5 = (1 + X + X^2) \cdot g(X)$
(0 0 0 1)	0 0 0 1 1 0 1	$X^3 + X^4 + X^6 = X^3 \cdot g(X)$
(1 0 0 1)	1 1 0 0 1 0 1	$1 + X + X^4 + X^6 = (1 + X^3) \cdot g(X)$
(0 1 0 1)	0 1 1 1 0 0 1	$X + X^2 + X^3 + X^6 = (X + X^3) \cdot g(X)$
(1 1 0 1)	1 0 1 0 0 0 1	$1 + X^2 + X^6 = (1 + X + X^3) \cdot g(X)$
(0 0 1 1)	0 0 1 0 1 1 1	$X^2 + X^4 + X^5 + X^6 = (X^2 + X^3) \cdot g(X)$
(1 0 1 1)	1 1 1 1 1 1 1	$1 + X + X^2 + X^3 + X^4 + X^5 + X^6$ $= (1 + X^2 + X^3) \cdot g(X)$
(0 1 1 1)	0 1 0 0 0 1 1	$X + X^5 + X^6 = (X + X^2 + X^3) \cdot g(X)$
(1 1 1 1)	1 0 0 1 0 1 1	$1 + X^3 + X^5 + X^6$ $= (1 + X + X^2 + X^3) \cdot g(X)$

Example

Obtain the parity-check polynomial. Construct a code polynomial to encode the message sequence 1001 using the generator polynomial shown in previous slide.

$$\text{(Sol)} \quad h(X) = \frac{X^7+1}{g(X)} = \frac{X^7+1}{1+X+X^3} = 1 + X + X^2 + X^4$$

Consider $m(X) = 1 + X^3$.

Multiply $m(X)$ by $X^{n-k} = X^3$, we get

$$X^{n-k}m(X) = X^3 + X^6$$

Next divide $X^{n-k}m(X)$ by $g(X)$:

Cont.

$$\begin{array}{r} X^3 + X \\ X^3 + X + 1 \overline{) X^6 + X^3} \\ \underline{X^6 + X^4 + X^3} \\ X^4 \\ \underline{X^4 + X^2 + X} \\ X^2 + X \end{array}$$

So we get $a(X) = X^3 + X$ and $b(X) = X^2 + X$

The desired code polynomial is

$$\begin{aligned} c(X) &= b(X) + X^{n-k}m(X) = X^2 + X + X^3 + X^6 \\ &= X + X^2 + X^3 + X^6 \end{aligned}$$

The codeword is, therefore, 0111001.

Generator Matrix

- For a given generator polynomial $g(X)$, we may construct the generator matrix of the code by noting that k polynomials $g(X), Xg(X), \dots, X^{k-1}g(X)$ span the code.
- Hence the n -tuples corresponding to these polynomials can be used as rows of the $k \times n$ generator matrix $\mathbf{G}$.
- Ex: Consider the generator polynomial $g(X) = 1 + X + X^3$. To construct the generator matrix of size 4×7 we start with the following polynomials:

$$\begin{aligned}g(X) &= 1 + X + X^3 \\Xg(X) &= X + X^2 + X^4 \\X^2g(X) &= X^2 + X^3 + X^5 \\X^3g(X) &= X^3 + X^4 + X^6\end{aligned}$$

Cont.

- The coefficients of the above polynomial are used as the rows of generator matrix:

$$\mathbf{G}' = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

- The generator matrix in systematic form is:

$$\mathbf{G} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Parity-Check Matrix

- The construction of parity-check matrix $\mathbf{H}$ requires special attention.
- As we know that $g(X)h(X) = X^n + 1$, multiplying on both sides with $a(X)$:

$$a(X)g(X)h(X) = X^n a(X) + a(X) \quad \Rightarrow \quad c(X)h(X) = a(X) + X^n a(X)$$

- $c(X)$ and $h(X)$ are polynomials of degree $n - 1$ and k respectively. So, in the above equation the left hand side contain terms with power extending up to $n + k - 1$.
- On the other hand, the polynomial $a(X)$ has degree $k - 1$ or less. $X^n a(X)$ contains the terms with powers from $X^n, \dots, X^{n-k+1}$. So, right hand side may have the powers from $0 \dots, k - 1, n, \dots, n - k + 1$.
- It means the powers $X^k, X^{k+1}, \dots, X^{n-1}$ are missing in the right hand side of the above equation.

Cont.

- Thus setting the coefficients of $X^k, X^{k+1}, \dots, X^{n-1}$ in the expansion of the product $c(X)h(X)$ equal to zero, we obtain the following set of $n - k$ equations:

$$\sum_{i=j}^{j+k} c_i h_{k+j-i} = 0 \text{ for } 0 \leq j \leq n - k - 1$$

- The above equation is as similar as $\mathbf{cH}^T = \mathbf{0}$.
- In the above equation, the coefficients of $h(X)$ are arranged in reverse order.
- This observation suggests that we define the reciprocal of the parity check polynomial as follows:

$$\begin{aligned} X^k h(X^{-1}) &= X^k \left(1 + \sum_{i=1}^{k-1} h_i X^i + X^k \right) \\ &= 1 + \sum_{i=1}^{k-1} h_{k-i} X^i + X^k \end{aligned}$$

Which is also a factor of $X^n + 1$.

Cont.

- The n -tuples pertaining to the $(n - k)$ polynomials $X^k h(X^{-1})$, $X^{k+1} h(X^{-1})$, ..., $X^{n-1} h(X^{-1})$ may now be used as rows of $(n - k) \times n$ parity-check matrix ***H***.
- Ex: Consider $h(X) = 1 + X + X^2 + X^4$. Obtain the three polynomials represented by $X^4 h(X^{-1})$ and two shifted version of it:

$$X^4 h(X^{-1}) = 1 + X^2 + X^3 + X^4 \quad \text{ROW 3}$$

$$X^5 h(X^{-1}) = X + X^3 + X^4 + X^5 \quad \text{ROW 2}$$

$$X^6 h(X^{-1}) = X^2 + X^4 + X^5 + X^6 \quad \text{ROW 1}$$

Cont.

- The coefficients of the above polynomial are used as the rows of parity-check matrix:

$$\mathbf{H}' = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

- The parity-check matrix in systematic form is: AFTER X OR OPERATION

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$